FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2023
FULL TEST – V
PAPER –1
TEST DATE: 08-02-2023

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

Section – A

1. B, D

Sol. $\frac{1000}{S_1} = \frac{2000}{S_2} \quad S_2 = 2S_1$

$S_1x + S_2y = (S_1 + S_2)40$

$S_1x + 2S_1y = (S_1 + 2S_1)40$

$x + 2y = 120$

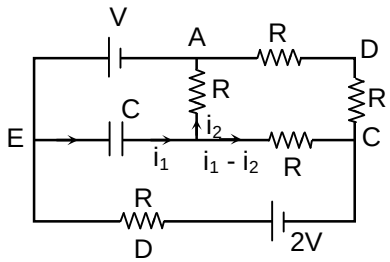
2. A, B, C

Sol. time constant of capacitor $(\tau) = \frac{5RC}{8}$

Maximum charge on capacitor at steady state $(q_0) = 5CV/4$

Charge on capacitor at any time

$q = \frac{5CV}{4}\left(1 - e^{-\frac{8t}{5RC}}\right)$ .... (i)



Current in the branch of capacitor at any time

$i_1 = \frac{2V}{R}e^{-\frac{8t}{5RC}}$

$i_1 = 1.7$ amp

Now, $v_A + i_2R(i_1 - i_2)R = V_C$

$V_A - v_C = 2.52$ volt

3. C, D

Sol. The force on the particle is not the central force.

4. A, B

Sol. For the disc (2m)

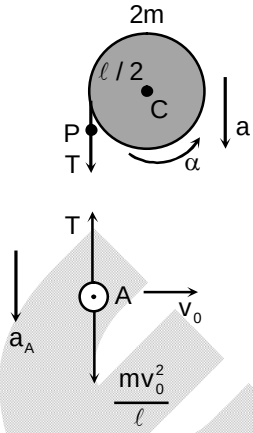
$$T = 2ma \quad ....(i)$$

$$T \times \frac{\ell}{2} = I_C \alpha \quad ....(ii)$$

For A (w.r.t. 'P')

$$T + m[a + R\alpha] = \frac{mv_0^2}{\ell}$$

Initial R.O.C. of $A = \dfrac{V_0^2}{(T/m)}$



5. A, B, C, D

Sol. $a_x + a_y = a_1$ ...(i)

$a_y - a_x = a_2$ ...(ii)

$a_{cm(x)} = 0$

$2a_2 + a_1 + a_x$ ...(iii)

6. A, B

Sol. $L = mv\theta r$ ...(i)

$$\frac{dL}{dt} = -\alpha v_0 r$$

$$\Rightarrow \frac{dL}{dt} = -\frac{\alpha L}{m} \quad \text{from (i)}$$

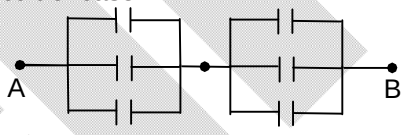
$$\Rightarrow \int_{L_0}^{L} \frac{dL}{dt} = -\frac{\alpha}{m}\int_0^t dt$$

$$L = L_0 e^{-\frac{\alpha t}{m}}$$

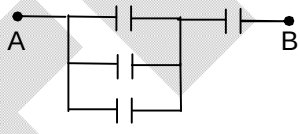
7. B

Sol. Eq. circuit of case P

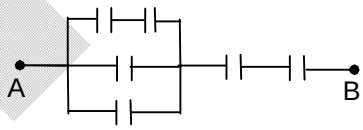
(P)



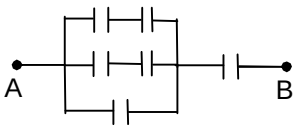
(Q)



(R)



(S)



8. A

Sol. (I) $C = C_v + C_v + \frac{RT}{v}\frac{dV}{dT} = C_v + \frac{Pdv}{dT}$ (for one mole gas)

we have $p = P_0 + \frac{\alpha}{v}$, or $\frac{RT}{v} = P_0 + \frac{\alpha}{v}$

or $RT = P_0v + \alpha$.

Therefore $RdT = P_0dv$, so, $\frac{dv}{dt} = \frac{R}{P_0}$

Hence $C = C_v + \left(P_0 + \frac{\alpha}{v}\right), \frac{R}{P_0} = \frac{R}{r-1} + \left(1 + \frac{\alpha}{P_0V}\right)R$

$= \left(R + \frac{R}{r-1}\right) + \frac{\alpha R}{P_0 v} = \frac{\gamma R}{\gamma - 1} + \frac{\alpha R}{P_0 v}$

(II) work done given by

$W = \int_{v_1}^{v_2}\left(P_0 + \frac{\alpha}{v}\right)^{dv} = P_0(v_2 - v_1) + a\ell n\frac{v_2}{v_1}$

(III) $\Delta U = C_v(T_2 - T_1) = C_v\left(\frac{P_2v_2 - P_1V_1}{R}\right) = \frac{1}{\gamma - 1}\left[p_2v_2 - p_1v_1\right]$

$= \frac{P_0(v_2 - v_1)}{\gamma - 1}$

(IV) $Q = DV + W = \frac{\gamma P_0(v_2 - v_1)}{\gamma - 1} + a\ln\frac{v_2}{v_1}$

9. D

Sol. (A) $(8 - 2t)\hat{j} - (8 - 2t)\hat{j}$

(B) $S = \frac{8 \times 8}{2 \times 2} = 16$

(C) $v = 8i - 8j, |v| = 8\sqrt{2}$

(D) $0 = 8t - \frac{1}{2}2t^2, t = 8$

10. B

Sol. If energy required by transition or ionization is Q then kinetic energy K required for colliding. Particle is

$$K = Q\left(1 + \frac{M_1}{M_2}\right)$$

here $M_1 = M_2$

$\therefore K = 2Q$

Minimum kinetic energy required for neutron to exite the atom = 2 × excitation energy.
Possible values of excitation energy = 10.2 ev, 12.1 ev,….
Ionisation energy = 13.6 ev

Section – B

11. 2.00

Sol. $\frac{\delta F}{\delta x} = k_{eff}\left(\text{numerically}\right)$

Let us find δx (static deformation of the system).

Since the particle is in equilibrium,

$\delta F = 2\delta F_s \sin\theta$

By Pythagoras theorem, $x^2 + y^2 = l^2$

Taking differentials of both sides, we have $2x\delta x + 2y\delta y = 2l\delta l$

Since y is constant, $\delta y = 0$

$\delta l = \frac{x}{l}\delta x$

$\frac{\delta F}{\delta x} = 2k\sin^2\theta$

$k_{eff} = 2k\sin^2\theta$

Substituting k_{eff}, we have $T = \frac{\pi}{\sin\theta}\sqrt{\frac{2m}{k}}$

12. 1.00

Sol. Velocity = $\omega\sqrt{A^2 - x^2}$

= $\frac{2\sqrt{3}\pi A}{T}$ (doubles)

$\frac{2\sqrt{3}\pi A}{T} = \frac{2\pi}{T}\sqrt{A^2 - x^2}$

$A_1 = \frac{\sqrt{13}}{2}A$

$\cos\theta = \frac{A/2}{A_1} = \frac{1}{\sqrt{13}} \quad \left(\theta = \cos^{-1}\left(\frac{1}{\sqrt{3}}\right)\right)$

$\cos^{-1}\left(\frac{1}{\sqrt{13}}\right) = \omega t$

$t = T\cos^{-1}\left(\frac{1}{\sqrt{13}}\right)$

13. 1.00

Sol. According to given condition,

$\frac{i\times 4}{i\times R_0} = \frac{AD}{DC} = \frac{40}{60} = \frac{2}{3}$

$\Rightarrow$ $R = 12$

Now, $R = R_0(1+\alpha\Delta T)$

$12 = 6(1+\alpha 100)$

$\Rightarrow \quad \alpha = 1\times10^{-2}$

So, $x = 1$

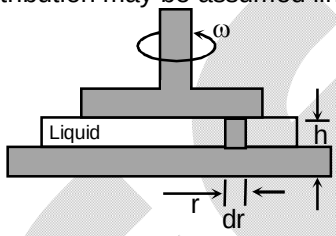
14. 4.00

Sol. Consider an element of disc at a radius r and having a width dr.

Linearly velocity at this radius = ωr

Shear stress, $\tau = \mu\dfrac{du}{dy}$

Assuming, the gap h to be small, so that the velocity distribution may be assumed linear



$\tau = \mu\times\dfrac{v}{h} = \mu\dfrac{\omega r}{h}$

Viscous force, $dF = \tau \times$ Area

$= \tau \times 2\pi rdr$

Torque dT on the element,

$dT = dF\times r = \tau 2\pi r^2 dr$

or $dT = \dfrac{\mu\omega r}{h}\times 2\pi r^2 dr = \dfrac{2\pi\mu\omega r^2 dr}{h}$

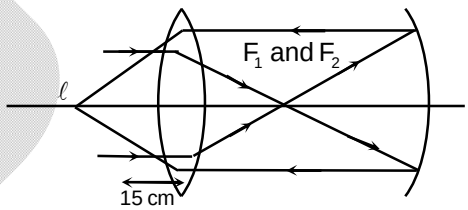
Total torque,

$$T = \int_0^{d/2}\frac{2\pi\mu\omega r^3 dr}{h} = \frac{\mu\pi d^4\omega}{4h}$$

Thus, x = 4

15. 4.00

Sol. The condition is possible only when the foci of lens and mirror coincides. Then, only the final image will be at 15 cm left to lens



So, $x = 15 + 21 = 36$ cm

Hence, $\dfrac{x}{9} = \dfrac{36}{9} = 4$ cm

16. 5.82

Sol. $y_{net} = y_1 + y_2 + y_3$

$= a[\sin\omega t + \sin(\omega t + 45°) + \sin(\omega t + 90°)]$

$= (\sqrt{2}+1)a\sin(\omega t + 45°)$

Energy ratio is

$$\frac{E_{resultant}}{E_{ring}} = \left(\frac{A}{a}\right)^2 = (\sqrt{2}+1)^2$$

$$= (3+2\sqrt{2})$$

$\therefore \quad E_{resultant} = 5.82$ J

17. 1.50

Sol. Thermal stress = $Y\alpha\Delta\theta$

As, thermal stresses are equal

$y_1\alpha_1\Delta\theta_1 = y_2\alpha_2\Delta\theta_2$

$\Rightarrow \quad \dfrac{Y_1}{Y_2} = \dfrac{\alpha_2}{\alpha_1} = \dfrac{3}{2}$

18. 2.00

Sol. Dimension of B = $[M^0L^0T^0]$

Dimension of e = [AT]

Dimension of $\varepsilon_0 = [A^2M^{-1}L^{-3}T^4]$

Dimension of h = $[ML^2T^{-1}]$

Dimension of c = $[LT^{-1}]$

Equating,

$$[M^0L^0T^0] = \frac{[AT]^n}{[A^2M^{-1}L^{-3}T^4][ML^2T^{-1}][LT^{-1}]}$$

$\Rightarrow$ $[M^0L^0T^0] = [A^{n-2}L^0T^{n-2}]$

$\therefore$ $n - 2 = 0$ or $n = 2$

Chemistry

PART – II

Section – A

19. B, D
Sol. Isotones have same number of neutrons

20. A, B, C, D
Sol. The $d_{A-A} = d_{B-B} = \sqrt{2}\, a/2$ and $d_{AB} = a/2$. There will be four molecules of AB in a unit cell. Hence. Its density will be given as $\rho = \frac{4 \times M_{AB}}{a^3 N_A}$.

21. B, D
Sol. RLVP = mole fraction of solute
Van't Hoff factor for CH_3COOH in benzene will be less than 1, because of association of CH_3COOH

22. A, B, C, D
Sol. Factual

23. A, B, C, D
Sol. Factual

24. B, D
Sol. (A) $CH_3 - CH = CH_2 \xrightarrow{H^+/H_2O} CH_3 - \underset{OH}{CH} - CH_3$

(B) $CH_3 - CH = CH_2 \xrightarrow[cold]{(aq)KMnO_4} CH_3 - \underset{OH}{CH} - \underset{OH}{CH_2}$

25. C
Sol. Factual

26. C
Sol. Factual

27. D
Sol. Factual

28. C
Sol. Factual

Section – B

29. 3.40
Sol. $I.P = R_H Z^2$, $13.6 = R_H$

$$\Delta E = R_H\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right) = 13.6\left(\frac{1}{2^2} - \frac{1}{\infty^2}\right) = 13.6 \times \frac{1}{4} = 3.4$$

30. 1.80

Sol. $\rho = \frac{PM}{RT}$ or $M = \frac{\rho \times RT}{P} = \frac{1.5 \times R \times 273}{1}$

$\rho = \frac{1000 \times 1.5 \times R \times 273}{760 \times R \times 300} = 1.8$

31. 181.14

Sol. for bcc, $\sqrt{3} \times a = 2(r_+ + r_-)$

$1.732 \times 387 = 2\ (154 + r_{Cl^-})$

$r_{Cl^-} \approx 181.14$

32. 6.67

Sol. $P_{PCl_5} = P_{Cl_3} = P_{Cl_2} = \frac{2}{6} \times 3 = 1$ atm

$K_p = \frac{1 \times 1}{1} = 1$

Let "y" mol of Cl_2 were added

	$PCl_5(g) \rightleftharpoons$	$PCl_3(g) +$	$Cl_2(g)$
Initial mol	2	2	$(2+y)$
mol of equilibrium	$(2+x)$	$(2-x)$	$(2+y-x)$

Total mol = 12, because total volume is twice of initial volume

Mol of $PCl_5 + PCl_3 = 2 + x + 2 - x = 4$

Mol of $Cl_2 = 12 - 4 = 8$

$PCl_5 = \frac{(2+x) \times 3}{12} = \frac{2+x}{4}$, $P_{PCl3} = \frac{2-x}{4}$, $P_{Cl_2} = \frac{8 \times 3}{12} = 2$

$\frac{\left(\frac{2-x}{4}\right) \times 2}{\frac{2+x}{4}} = 1.0 \Rightarrow x = \frac{2}{3}$

$2 + y + \frac{2}{3} = 8$

$y = \frac{20}{3} = 6.67$

33. 300.00

Sol. For $Ag_2C_2O_4 \rightleftharpoons 2Ag^+ + C_2O_4^{2-}$

$S = \left(\frac{K_{sp}}{4}\right)^{\frac{1}{3}} = \left(\frac{1.35 \times 10^{-11}}{4}\right)^{\frac{1}{3}} = \left(3.375^{-12}\right)^{\frac{1}{3}} = 1.5 \times 10^{-4}$

$\Lambda_m = \frac{k \times 1000}{S}$

$\Lambda_m = \frac{4.5 \times 10^{-5} \times 10^3}{1.5 \times 10^{-4}} = 300$ S cm^2 mol^{-1}

34. 138.50

Sol. $X_{H_2O} = 1 - \frac{1}{7} = \frac{6}{7}$

$$\frac{X_{KClO_4}}{X_{H_2O}} = \frac{1}{7} \times \frac{7}{6} = \frac{1}{6} = \frac{n(KClO_4)}{n(H_2O)}$$

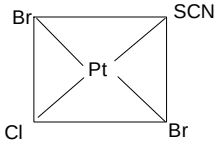
It mean 6 mol H_2O (108g H_2O) contain 1 mol $KClO_4$ = 138.5g $KClO_4$

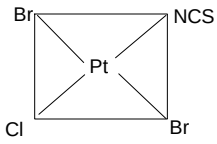
35. 6.10

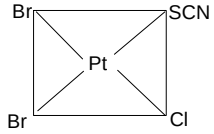
Sol. Factual

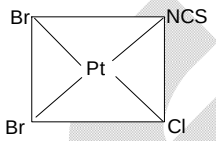
36. 4.00

Sol.









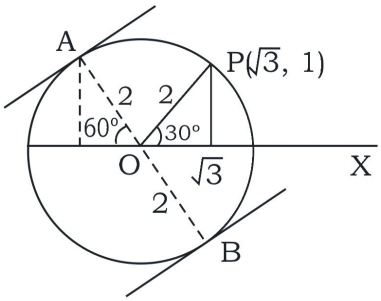
Mathematics

PART – III

Section – A

37. B, C

Sol.

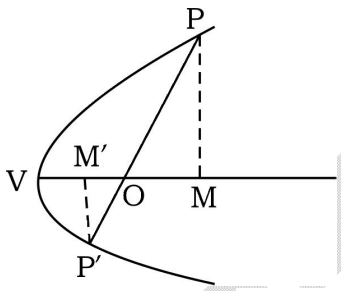


Clearly, $A=\left(-2\cos 60^{\circ}, 2\sin 60^{\circ}\right)$ *and* $B=\left(2\cos 60^{\circ}, -2\sin 60^{\circ}\right)$.

The tangent at A is $x\left(-2\cos 60^{\circ}\right)+y\left(2\sin 60^{\circ}\right)=4$ and the tangent at B is $x\left(2\cos 60^{\circ}\right)+y\left(-2\sin 60^{\circ}\right)=4$.

38. B

Sol.



Let the parabola be $y^2=4ax$ and $p=\left(at_1^{\,2}, 2at_1\right), P'=\left(at_2^{\,2}, 2at_2\right)$

$\therefore \quad VM=at_1^{\,2}, VM'=at_2^{\,2} and\ VO=k, where \begin{vmatrix} a_1^2 & 2at_1 & 1 \\ a_2^2 & 2at_2 & 1 \\ k & 0 & 1 \end{vmatrix} = 0$

$(\because \quad p, p', O\ are\ collinear)$

$\Rightarrow \quad k+at_1t_2=0$

$\therefore \quad VM\cdot VM'=\left(at_1t_2\right)^2=k^2=OV^2.$

39. B, C

Sol. $\alpha=2\tan^{-1}\left(\sqrt{2}-1\right)=2\tan^{1}\tan\frac{\pi}{8}=2\times\frac{\pi}{8}=\frac{\pi}{4}=\cos^{-1}\frac{1}{\sqrt{2}}.$

$\beta=3\cdot\frac{\pi}{4}-\frac{\pi}{6}=\frac{7\pi}{12}.$

$\therefore \quad \beta>\alpha\cdot Also\ \frac{1}{3}<\frac{1}{\sqrt{2}} \quad \Rightarrow \cos^{-1}\frac{1}{3}>\cos^{-1}\frac{1}{\sqrt{2}}=\frac{\pi}{4}.$

So $\gamma>\alpha.$

Again $\cos^{-1}\frac{1}{3}$ belongs to the first quadrant and β is in the second quadrant.

$\therefore \quad \beta > \gamma.$

40. A, B, D

Sol. $\int_4^8 \left(\left|\sqrt{x-4}+2\right|+\left|\sqrt{x-4}-2\right|\right)dx = \int_4^8 4dx = 16$

Similarly option A = B = D = 16

41. B, C, D

Sol. $\det(M-I) = \det(M-I)^T$

$= \det(M^T - I)$

$= \det(-M-I) = \begin{cases} \det(M+I), & \text{if } n \text{ is even} \\ -\det(M+I), & \text{if } n \text{ is odd} \end{cases}$

42. A, B

Sol. Since, $\vec{a}$ and $\vec{b}$ are unit vectors, we have

$\left|\vec{a}-\vec{b}\right| = \sqrt{\left(\vec{a}-\vec{b}\right)^2}$

$\therefore \sqrt{\left(\vec{a}\right)^2+\left(\vec{b}\right)^2-2\vec{a}.\vec{b}} = \sqrt{1+1-2\cos 2\theta} = 2\left|\sin\theta\right|$

Therefore, $\left|\vec{a}-\vec{b}\right| < 1$

$\Rightarrow \quad 2\left|\sin\theta\right| < 1$

$\left|\sin\theta\right| < \frac{1}{2}$

$\Rightarrow \quad \theta \in \left[0, \frac{\pi}{6}\right) \quad \text{or} \quad \left(\frac{5\pi}{6}, \pi\right]$

43. B

Sol. I.

$OA = 1 + 4\cot\theta$

$OB = 4 + \tan\theta$

$OA + OB = 5 + 4\cot\theta + \tan\theta \geq 5 + 2\sqrt{4\cot\theta\tan\theta}$

$= 5 + (2\times 2) = 9$

II.

The reflection of P(4, –1) on y = x is Q (–1, 4).

Hence, PQ = $\sqrt{(4+1)^2(-1-4)^2} = \sqrt{50} = 5\sqrt{2}$.

III. AB = $2\sqrt{2}$

OC = $\sqrt{2}$

The maximum value of d is

OF = $\sqrt{2} + 2\sqrt{2} = 3\sqrt{2}$

IV.

The given line is $x = 4 + \frac{1}{\sqrt{2}}\left(\frac{y+1}{\sqrt{2}}\right)$ or $y = 2x - 9$

Hence, the intercept made by the x-axis is 9/2.

44. C

Sol. I. We have $\frac{1}{\log_3 \pi}+\frac{1}{\log_4 \pi}=\log_\pi 3+\log_\pi 4=\log_\pi 12$

But $\pi^2<12<\pi^3$, we have $2 < \log_\pi 12 < 3$.

II. $3^a = 4; a = \log_3 4$

Similarly, $b = \log_4 5$ etc.

Hence, $abcdef = \log_3 4 \cdot \log_4 5 \cdot \log_5 6 \cdot \log_6 7 \cdot \log_7 8 \cdot \log_8 9 = \log_3 9 = 2$

III. We have to find characteristic of $\log_2 2008$.

We know that $\log_2 1024 = 10$ and $\log_2 2048 = 11$, therefore $10 < \log_2 2008 < 11$

Hence, it has characteristic 10.

IV. $\log_2\left(\log_2\left(\log_3 x\right)\right)=0 \Rightarrow \log_2\left(\log_3 x\right)=1 \Rightarrow \log_3 x = 2$

$\Rightarrow x = 9$

Similarly, we have $\log_3\left(\log_2 y\right)=1$

$\Rightarrow \log_2 y = 3$ or $y = 8$

Therefore, $x - y = 1$.

45. A

Sol. I. $I=\lim_{x\to 0}\frac{e^{x^2}-1+x-e^x+1}{2\frac{\sin^2 x}{x^2}.x^2}=\frac{1}{2}\left[\lim_{x\to 0}\frac{e^{x^2}-1}{x^2}+\lim_{x\to 0}\frac{x-e^x+1}{x^2}\right]=\frac{1}{2}\left[1-\lim_{x\to 0}\frac{e^x-x-1}{x^2}\right]$

II. $I=\lim_{x\to 0}\left(\frac{3+x}{3-x}\right)^{1/x}=e^{\lim_{x\to 0}\frac{1}{x}\left(\frac{3+x}{3-x}-1\right)}=e^{\lim_{x\to 0}\frac{2x}{x(3-x)}}=e^{2/3}\Rightarrow 2+3=5$

III. $\lim_{x\to 0}\frac{\left(\tan^3 x-x^3\right)-\left(\tan x^3-x^3\right)}{x^5}$

$=\lim_{x\to 0}\frac{\tan^3 x-x^3}{x^5}-\underbrace{\lim_{x\to 0}\frac{\tan^3 x-x^3}{x^5}}_{\text{zero(by expansion)}}$

$==\lim_{x\to 0}\frac{(\tan x-x)}{x^3}.\frac{\left(\tan^2 x+x\tan x+x^2\right)}{x^2}=\frac{1}{3}\times 3=1$

IV. Rationalising given

$\lim_{x\to 0}\frac{(x+2\sin x)\left[\sqrt{(x^2+2\sin x+1)}+\sqrt{\sin^2-x-x+1}\right]}{(x^2+2\sin x+1)-(\sin^2 x-x+1)}$

$=2.\lim_{x\to 0}\frac{x+\sin 2x}{x^2-\sin^2 x+2\sin x+x}$

$=2.\lim_{x\to 0}\frac{1+\frac{\sin 2x}{x}}{x-\frac{\sin^2 x}{x}+2+1}=2\left(\frac{1+2}{3}\right)=2$

46. C

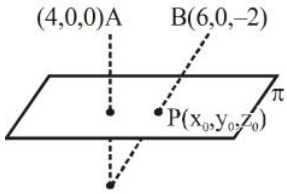
Sol. Let the plane be $\alpha x + \beta z + 1 = 0$

Pass through (1, 0, 1); (3, 2, –1)

$\therefore \alpha=-\frac{1}{2};\ \beta=-\frac{1}{2}$

(I) $\pi : x + z = 2$

(II)



Both A and B are on same side of π.
Reflection of A in plane is A'(2, 0, –2)
Equation of line A'B = $\vec{r} = 6\hat{i} - 2\hat{k} + \lambda(4\hat{i})$

For P : $6 + 4\lambda + 0 - 2 = 2 \Rightarrow \lambda = \frac{-1}{2}$

$\therefore$ P(4, 2, – 2)

$\therefore \ |4x_0 + y_0 + 2z_0| = 12$

(III) Now $|PA - PB|_{min} = 0$
$|PA - PB|_{min}$ will approach
AB = $\sqrt{4+0+4} = \sqrt{8}$
$\therefore |PA - PB| \in [0, \sqrt{8})$.

(IV) Also, A' will lie on $\frac{x-2}{1} = \frac{y-\alpha}{0} = \frac{z+\beta}{-1}$.

$\Rightarrow \frac{2-2}{1} = \frac{0-\alpha}{0} = -\frac{2+\beta}{-1}$

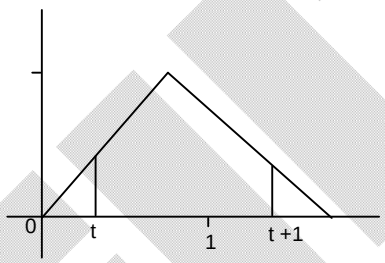
$\Rightarrow \alpha = 0, \beta = 2$

$\therefore \alpha^4 + \beta^4 = 16$

Section – B

47. 2.00

Sol. The region M is the large triangle in the diagram below, and the region N is the infinite strip between the lines $x = t$ and $x = t + 1$where $0 < t < 1$



Hence $M \cap N$ is the pentagon. Its area is given by

$$\frac{1}{2}\left(\sqrt{2}^2 - t^2 - \left(2 - (t+1)\right)^2\right) = -t^2 + t + \frac{1}{2}$$

48. 6.00

Sol. $t^2 + 3t - 8 = 2 \quad \Rightarrow t = 2, -5$

$2t^2 - 2t - 5 = -1 \Rightarrow t = 2, -1$

So, at $t = 2$

$$\frac{dy}{dx} = \left(\frac{\frac{dy}{dt}}{\frac{dx}{dt}}\right)_{t=2} = \frac{6}{7}$$

49. 4320.00

Sol. a_3 is coefficient of x^3

$$\frac{10!}{8!}\times 3\times 4+\frac{10!}{7!3!}\times 3^3=4320$$

50. 1.00

Sol. Applying $C_1\to C_1-\sin\theta C_3$ and $C_2\to C_2+\cos\theta C_3$, we get

$$f(\theta)=\begin{vmatrix}1 & 0 & -\sin\theta\\ 0 & 1 & \cos\theta\\ \sin\theta & -\cos\theta & 0\end{vmatrix}$$

Applying $R_3\to R_3-\sin\theta R_1+\cos\theta R_2$, we get

$$f(\theta)=\begin{vmatrix}1 & 0 & -\sin\theta\\ 0 & 1 & \cos\theta\\ \sin\theta & 0 & \sin^2\theta+\cos^2\theta\end{vmatrix}=1$$

Thus, $f\left(\frac{\pi}{6}\right)=1$

51. 7.00

Sol. We have

$$(\alpha+\beta+\gamma)^2=\alpha^2+\beta^2+\gamma^2+2(\alpha\beta+\beta\gamma+\gamma\alpha)$$

$$\Rightarrow \quad 16=6+2(\alpha\beta+\beta\gamma+\gamma\alpha)$$

$$\therefore \quad \alpha\beta+\beta\gamma+\gamma\alpha=5$$

Also, $\alpha^3+\beta^3+\gamma^3-3\alpha\beta\gamma=(\alpha+\beta+\gamma)(\alpha^2+\beta^2+\gamma^2-\alpha\beta-\beta\gamma-\gamma\alpha)$

$$\Rightarrow \quad 8-3\alpha\beta\gamma=4(6-5)$$

or $3\alpha\beta\gamma=4$

or $\alpha\beta\gamma = 4/3$

Now, $(\alpha^2+\beta^2+\gamma^2)^2=\sum\alpha^4+2\Sigma\beta^2\gamma^2$

$$=\sum\alpha^4+2\left\{(\Sigma\beta\gamma)^2-2\alpha\beta\gamma\sum\alpha\right\}$$

$$(6)^2=\sum\alpha^4+2\left\{25-2.\frac{4}{3}.4\right\}$$

$$\sum\alpha^4=36-50+\frac{64}{3}=\frac{64}{3}-14=\frac{22}{3}$$

$$\therefore \quad \left[\alpha^4+\beta^4+\gamma^4\right]=\left[\frac{22}{3}\right]=7$$

52. 0.00

Sol. Coefficient of x^{103} in $(1 + x + x^2 + x^3 + x^4)^{199} (x - 1)^{201}$

Coefficient of x^{103} in $-(1 - x)^2 (1 - x^5)^{199}$

Coefficient of x^{103} in $-(1 - 2x + x^2) (1 - {}^{199}C_1 x^5 + {}^{199}C_2 x^{10} - {}^{199}C_3 x^{15} + ...)$

Coefficient of $x^{103} = 0$

53. 2022.00

Sol. For $x > 0$ not possible

For $x < 0$ put $x = -y$

$$\left(1-\frac{1}{y}\right)^{1-y} = \left(\frac{y-1}{y}\right)^{1-y} = \left(\frac{y}{y-1}\right)^{1-y} = \left(\frac{2022}{2021}\right)^{2021}$$

$\Rightarrow y = 2022$

$x = -2022$

$|x| = 2022$

54. 0.25

Sol. $P(B)\ (A \cup \bar{B})) = \dfrac{P(B \cap (A \cup \bar{B})}{P(A \cup \bar{B})} = \dfrac{P(A) - P(A \cap \bar{B})}{P(A) + P(\bar{B}) - P(A \cap \bar{B})}$

$$= \frac{0.7-0.5}{0.7+0.6-0.5} = \frac{0.2}{0.8} = \frac{1}{4}$$